

The Fertility, Marriage, and Gender Equality Quandary

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Motivation

- Three trends underpin the grand gender convergence in the last century:
 1. Falling fertility (Guinnane 2011)
 2. Declining dual parenthood (Stevenson and Wolfers 2007)
 3. Converging gender (income) gaps (Goldin 2014)

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- Many policies target individual outcomes
 1. Incentivize fertility – pension sustainability, growth. . .
 2. Promote dual parenthood – religion/culture, children's outcomes. . .
 3. Advance gender equality – redistribution, reducing misallocation. . .

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- Existing research treats these outcomes in isolation or pairwise, lacking a unified framework

This Paper

1. Document a **three-way trade-off**: simultaneously achieving high fertility, low single parenthood, and gender income equality is statistically rare
⇒ deep dependency structure, potential policy boundaries
2. Develop a **equilibrium model** of marriage, fertility, and female labor supply
3. Show level policies improve two at the expense of the third; **redistributing childcare** can loosen the trade-off under transparent conditions
4. Quantitative analysis and dynamic extension

Motivating Facts

Data Construction

	OECD	World	U.S. States	U.S. CZ
Coverage	37 countries	95 countries	50 + D.C.	656 CZs
Period	1970–2014	1990–2015	2023	2000
Obs.	721	1,130	51	656
Fertility	TFR	TFR	Births / 1000	TFR
Dual parenthood	Out-of-wedlock	Living arrangement	Single-mother	Single-mother
Gender gap	Earnings	Income share	Earnings	Earnings
Advantage	Long horizon	Wide coverage	Uniform inst.	Granular

- Construct dummies at sample medians: high F, dual M, equal G
- **Dartboard benchmark**: under independence, trinity share is $0.5^3 = 12.5\%$

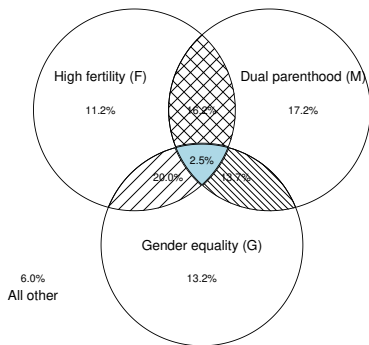
The Trinity is Rare

OECD

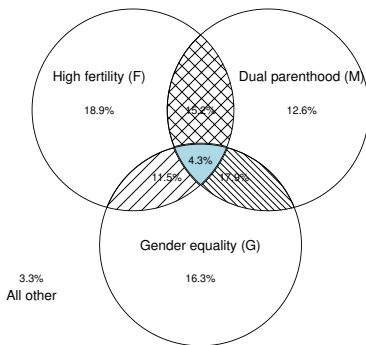
US States

US CZ

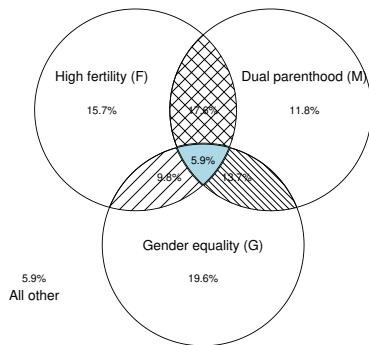
synergy



OECD



World



U.S. States

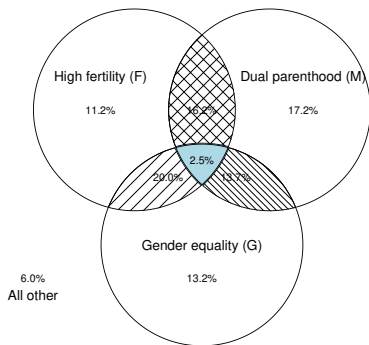
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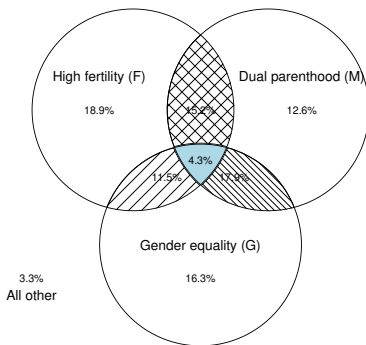
US States

US CZ

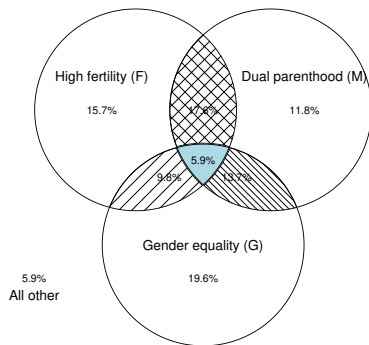
synergy



OECD



World



U.S. States

1. Typical transition: losing F and/or M, gaining G
2. Mutual information analysis shows synergy
3. Results are robust across income and education groups

Questions

1. What explains the three-way trade-off?
2. Are there policies that can resolve the trade-off?
3. What role does technological progress play?

Theory

Basic Setup

- Building on Becker (1973), Choo and Siow (2006)
- Individuals with gender $g \in \{\sigma, \varphi\}$ and preference

$$u(c, n) = \left((1 - \beta) \cdot c^{\frac{\rho-1}{\rho}} + \beta \cdot n^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}} \quad (1)$$

where $\rho > 1$ so that $u(c, 0)$ is well-defined

- Homogeneous wage within gender: w^σ and w^φ
- Gender wage gap: $\Gamma^w = w^\sigma / w^\varphi$

Marriage and Fertility – Men

- If single, men consume labor income but have no children

$$V^{\sigma^{\rightarrow},s} = u(w^{\sigma^{\rightarrow}}, 0) \quad (2)$$

- Once married, husbands transfer α share of income to wives

$$V^{\sigma^{\rightarrow},m} = u((1 - \alpha)w^{\sigma^{\rightarrow}}, n^m) \cdot \lambda \quad (3)$$

- $\alpha \in [-w^{\sigma^{\rightarrow}}/w^{\sigma^{\rightarrow}}, 1]$ is endogenous in marriage market equilibrium
- λ is the psychic benefit of marriage

Marriage and Fertility – Women

- Single women solve

$$V^{\text{♀},s} = \max_{c,l,n} u(c, n) \quad \text{s.t.} \quad c = w^{\text{♀}}l, \quad l = 1 - \chi n \quad (4)$$

- Married women solve

$$V^{\text{♀},m} = \max_{c,l,n} u(c, n) \cdot \lambda \quad \text{s.t.} \quad c = \alpha w^{\text{♂}} + w^{\text{♀}}l, \quad l = 1 - \chi n \quad (5)$$

- Trade-off between consumption and fertility through female labor supply
- In the baseline, wives bear all childcare. Results hold so long as $\chi^{\text{♂}} < \chi^{\text{♀}}$

Marriage Market Equilibrium

- Idiosyncratic taste shock ϵ for marriage, Fréchet with scale θ
- Marriage rates:

$$\mathcal{M}^{\sigma} = \frac{1}{1 + (V^{\sigma,s}/V^{\sigma,m})^{\theta}}, \quad \mathcal{M}^{\varphi} = \frac{1}{1 + (V^{\varphi,s}/V^{\varphi,m})^{\theta}} \quad (6)$$

- Equilibrium: $\mathcal{M}^{\sigma} = \mathcal{M}^{\varphi} = \mathcal{M}$
- Two-dimensional price (α, n^m) : marriage bundles income transfer *and* marital fertility

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- **Two-dimensional price** (α, n^m) : marriage bundles income transfer *and* marital fertility
- Lemma 1: For any $\{w^{\sigma}, w^{\varphi}\}$, there exists a unique equilibrium (α, n^m)
- Lemma 2: Gains from marriage: $V^{\varphi,m}/V^{\varphi,s} = \lambda \cdot (1 + \alpha\Gamma^w)$

Three-Way Trade-Off: Intuition

- Equilibrium conditions reveal fundamental tension:

$$\mathcal{M} = \frac{(\lambda \cdot (1 + \alpha\Gamma^w))^\theta}{1 + (\lambda \cdot (1 + \alpha\Gamma^w))^\theta} \quad (7)$$

$$\Gamma^y = \Gamma^w / l^\varnothing \quad (8)$$

$$l^\varnothing = 1 - \chi n \quad (9)$$

- Insight:** Higher n reduces $l^\varnothing$, requiring low Γ^w for gender income equality—but low Γ^w reduces $\mathcal{M}$

Traditional Policy Effects

Policy	Fertility (n)	Marriage ($\mathcal{M}$)	Gender Gap (Γ^y)
Pro-marriage ($\uparrow \lambda$)	+	+	+ (widens)
Gender equality ($\uparrow w^{\text{♀}}$)	−	−	− (narrows)
Reduce child costs ($\downarrow \chi$)	+	+	+ (widens)

- No *level* policy can improve all three outcomes
- Trade-off binds because
 1. Wives' disproportionate childcare burden ties fertility to gender gaps through relative labor supply
 2. Exchange motive and composition effect ($n^m > n^s$) ties fertility to marriage

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 1. Wives' disproportionate childcare burden ties fertility to gender gaps through relative labor supply
 2. Exchange motive and composition effect ($n^m > n^s$) ties fertility to marriage
- What if the **allocation** of childcare is itself a policy lever?

Model with Husband-Provided Childcare

- Extension: $\chi^{\text{♀}}$ for wives, $\chi^{\text{♂}}$ for husbands
- Social norm as a reason for exogenous $(\chi^{\text{♀}}, \chi^{\text{♂}})$ that may be inefficient
- $\chi = \chi^{\text{♀}} + \chi^{\text{♂}}$ for single mothers
- Wife's effective childcare cost: $\tilde{\chi}^{\text{♀}} = \frac{\alpha w^{\text{♂}} \chi^{\text{♂}} + w^{\text{♀}} \chi^{\text{♀}}}{\alpha w^{\text{♂}} + w^{\text{♀}}}$
- All previous results go through as long as $\chi^{\text{♀}} > \chi^{\text{♂}}$
- Consider redistributing δ childcare from wives to husbands

Redistributing Childcare to Husbands

- **Proposition:** Redistribution improves all three outcomes if:

1. Wife's earnings exceed spousal transfers:

$$w^{\varnothing} > \alpha \cdot w^{\sigma} \quad (10)$$

2. Husband's gain from fertility exceeds consumption loss.

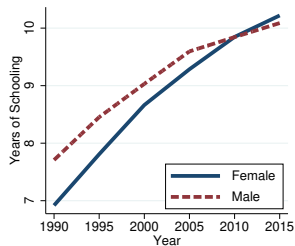
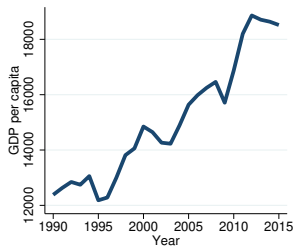
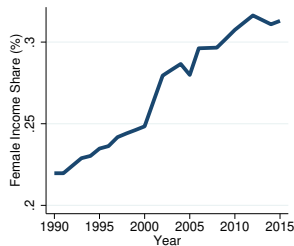
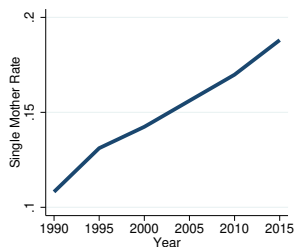
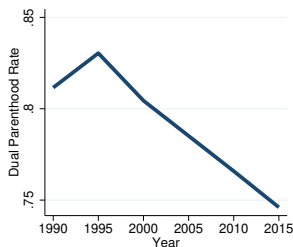
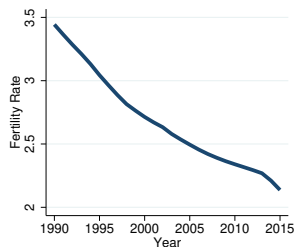
$$\frac{\beta}{1-\beta} \cdot \left(\frac{c^{\sigma,m}}{n^m} \right)^{\frac{1}{\rho}} > (1-\alpha)w^{\sigma} \cdot \left(\frac{n^m}{\partial n^m / \partial \delta} + \chi^{\sigma} \right), \quad (11)$$

More likely if: β is large, initial χ^{σ} is small, and fertility response is strong

- When both conditions hold, redistribution is **Pareto-improving**

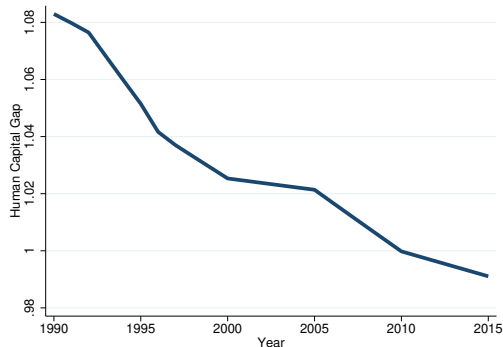
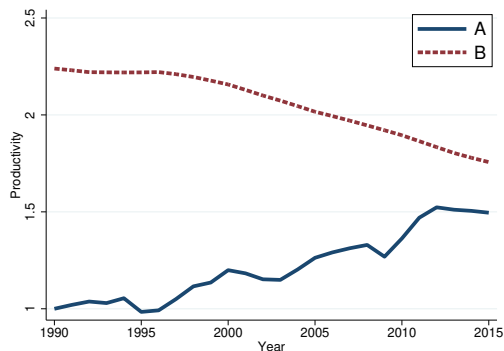
Quantitative Analysis

Mexico's Transition (1990–2015)



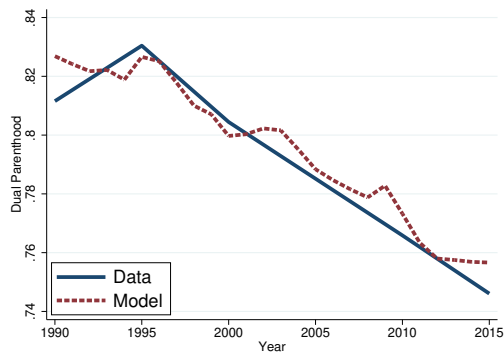
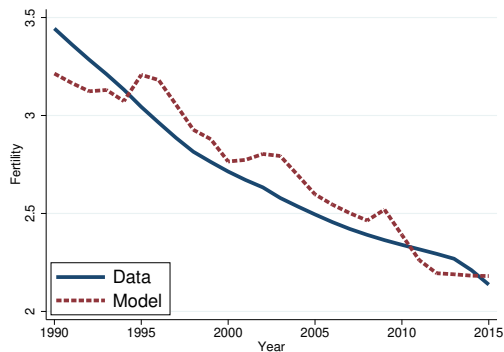
Calibration

- Decompose wages into three structurally interpretable components:
 - $A_t = w_t^{\sigma}$: gender-neutral productivity
 - Γ_t^h : gender gap in human capital
 - $B_t = (w_t^{\sigma} / w_t^{\varphi}) / \Gamma_t^h$: gender-biased productivity



Model Fit

- Fix $\chi = 0.075$ (de La Croix and Doepke 2003), assuming $\chi^{\sigma^*} \approx 0$
- Calibrate: $\beta = 0.104$, $\rho = 1.50$, $\lambda = 1.03$, $\theta = 3.40$



Key Quantitative Results

1. Stable α_t despite falling n_t^m because narrowing Γ_t^w requires a larger share of husbands' falling relative income
2. Gender-neutral productivity growth explains about half of the decline in fertility and marriage
3. Gender-biased productivity growth and schooling gap reversal accounts for the convergence in gender welfare gaps

Dynamic Extension

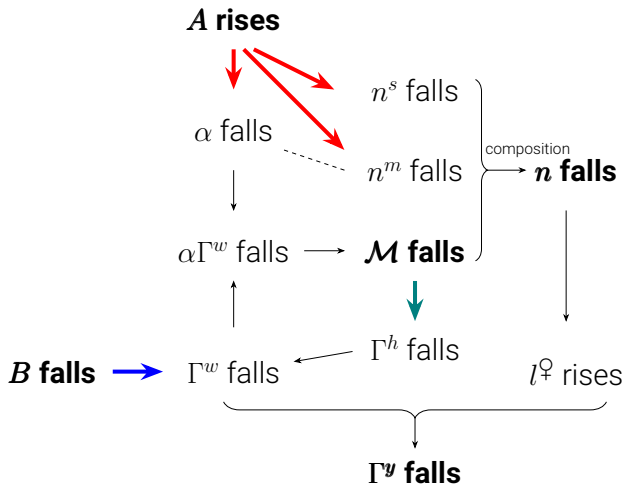
Endogenous Gender Human Capital Gap

- Single parenthood affects boys and girls differently
- Evidence: Bertrand & Pan 2013, Autor et al. 2019, Wasserman 2020, Reeves 2022, Autor et al. 2023
- Law of motion:

$$\Gamma_{t+1}^h = \mathcal{H}(\mathcal{M}_t) \quad \text{where } \mathcal{H}'(\mathcal{M}_t) > 0 \quad (12)$$

- Creates self-reinforcing feedback: declining marriage $\rightarrow$ lower male human capital $\rightarrow$ further marriage decline

Dynamic Mechanism



Red: A_t rises Blue: B_t falls Teal: Dynamic feedback

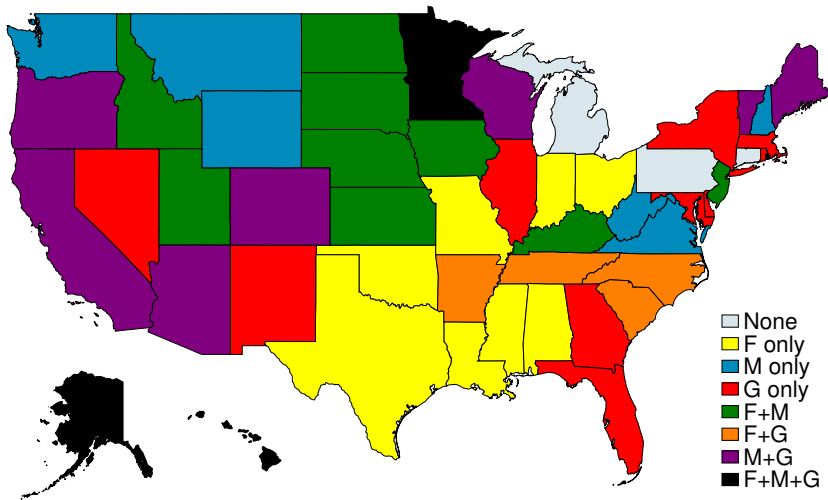
Conclusion

- A three-way trade-off with synergistic dependency structure
- Unified framework with two-dimensional marriage-market pricing
- Level policies trade one outcome against the others, while redistribution of childcare can loosen the trade-off under checkable conditions
- Mexico: both gender-neutral and gender-biased technological change matter for the transition
- Dynamic propagation through male human capital amplifies effects across generations

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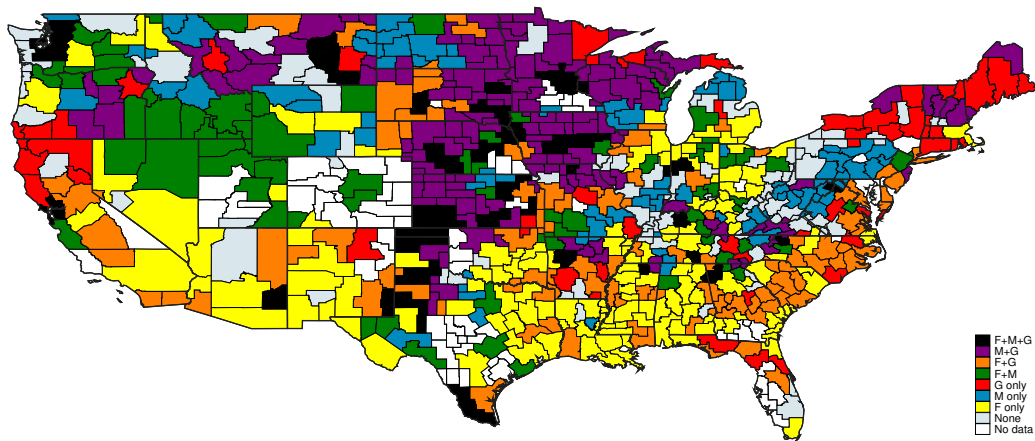
U.S. States – Geographic Distribution

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Three states achieve the trinity: [Alaska](#), [Hawaii](#), [Minnesota](#)

U.S. CZ – Geographic Distribution

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Synergy, not Redundancy

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- **Synergy**: conditioning on any one variable *strengthens* the dependence between the other two
- Implication: pairwise analyses miss joint structure

Sample	Total Corr.	Interaction	Interpretation
OECD	0.165	−0.035***	Synergy
World	0.169	−0.026***	Synergy
U.S. States	0.112	−0.007	Synergy

Example (OECD): $I(F; M) = 0.043$ bits, but $I(F; M | G) = 0.078$ bits—**nearly double**